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Time-Domain Heart Rate Variability as a Practical Tool to Assess Readiness and Recovery in Youth Soccer Athletes

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Abstract

Heart rate variability (HRV) is a non-invasive biomarker widely used to assess autonomic regulation, training adaptation, and recovery status in athletes. However, studies involving soccer players under 17 years of age remain scarce, particularly those integrating physiological and psychophysiological markers in real competitive contexts. This study aimed to analyze autonomic modulation using time-domain HRV (natural logarithm of the root mean square of successive difference LnRMSSD) and its relationship with subjective indicators of recovery and load in under-17 elite soccer athletes. An observational longitudinal study was conducted with fourteen elite male soccer players from Santos Futebol Clube (age: 16.2 $\pm$ 0.42 years) monitored over four competitive weeks. Morning HRV was recorded using a validated ultra-short two-minute protocol, while psychophysiological measures included perceived recovery status (PRS), session rating of perceived exertion (sRPE), perceived stress, and mood. The athletes demonstrated no statistically significant temporal changes in RMSSD or LnRMSSD throughout the monitoring period, with mean values of LnRMSSD = 4.06 $\pm$ 0.68 and RMSSD = 68.78 $\pm$ 33.62 ms. Significant associations were observed between LnRMSSD and PRS (r = 0.61; $p < 0.04$), sRPE (r = -0.67; $p < 0.03$), and perceived stress (r = -0.77; $p < 0.01$), indicating, under the conditions of this study, coherence between physiological and perceptual indicators of internal load and recovery status. In conclusion, the results suggest that elite under-17 soccer players maintained stable autonomic modulation during the competitive period analyzed. Furthermore, the observed associations between HRV and subjective measures support the potential value of integrating physiological and perceptual indicators when monitoring recovery and readiness in youth soccer athletes.

Keywords: heart rate variability; autonomic nervous system; recovery status; training load monitoring

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1. Introduction

Heart rate variability (HRV) represents the temporal fluctuation between consecutive heartbeats and reflects the dynamic interaction between the sympathetic and parasympathetic branches of the autonomic nervous system (ANS) [1]. Because HRV is strongly

influenced by vagal modulation, it has been widely used as a non-invasive marker of autonomic regulation, recovery status, and adaptive capacity in response to physical and emotional stressors [2,3]. In athletes, HRV has been progressively applied to monitor internal load, readiness, and recovery, particularly in high-performance soccer contexts [4,5].

HRV can be quantified using different parameters, such as time domain, frequency domain, and through nonlinear methods [6]. However, because it is a non-invasive, inexpensive and highly efficient method in the detection of autonomic balance, mediated by vagal modulation under the parasympathetic system of athletes, the use of the time domain parameter has been the object of research in recent studies with soccer athletes [7]. The calculated RMSSD (Root Mean Square of Successive Differences) is a time-domain measure of heart rate variability (HRV) that quantifies the short-term variability between consecutive normal-to-normal (NN) heartbeats. It is calculated as the square root of the mean of the squared differences between successive NN intervals. RMSSD predominantly reflects parasympathetic (vagal) modulation of cardiac activity and is commonly used to assess autonomic regulation, recovery status, and physiological responses to training. RMSSD is especially responsive to short-term autonomic changes associated with physiological recovery and neuromuscular readiness in athletes [6,8].

In team sports, HRV monitoring is particularly useful because it is time-efficient, non-invasive, and feasible within the complex daily routines of training and competition, allowing practitioners to integrate physiological and psychophysiological information into load-monitoring strategies [9,10].

Studies examining changes in heart rate variability (HRV) across different phases of the competitive season or during seasonal adaptation periods have consistently demonstrated patterns of parasympathetic suppression during periods of elevated physical and psychological load.

For instance, Flatt et al. [11] observed, in a longitudinal study with collegiate American football players, a progressive reduction in LnRMSSD (natural logarithm of the root mean square of successive difference, a time-domain heart rate variability index derived from the differences between consecutive normal RR intervals) values throughout the season, particularly among athletes with higher body mass, suggesting autonomic imbalance associated with accumulated training load and physiological stress.

More recently, Lipka et al. [12] found that multiple studies involving professional soccer players reported associations between linear HRV indices (time and frequency domains) and markers of overtraining, physical performance, fatigue, and endocrine responses. These findings reinforce the utility of HRV as a reliable indicator of internal training load and physiological adaptation. Tekin et al. [13] demonstrated that, among football athletes, resting HRV correlates not only with aerobic capacity but also with agility, neuromuscular coordination, sleep quality, and body awareness, broadening the scope of autonomic monitoring beyond basic physical conditioning.

Furthermore, O'Connor et al. [14] have shown that fluctuations in internal training load across 1, 3, and 7-day periods predict changes in morning HRV among elite athletes, supporting the concept of cumulative autonomic responses over the competitive season.

Although HRV is widely used in elite professional soccer [15], evidence in youth soccer remains limited. This population is exposed not only to high physical demands but also to emotional, psychosocial, and routine-related factors that may influence autonomic regulation and recovery [16–18]. In addition, individual factors such as sleep quality, perceived exertion, nutrition, and supplementation may modulate vagal responses and should be considered when interpreting HRV data in young athletes [19,20].

Few studies, however, have examined HRV in athletes under 17 years of age by means of longitudinal monitoring that integrates psychophysiological variables with methodolog-

ically rigorous protocols. Considering the scarcity of data on elite youth soccer and its contextual complexity, it is necessary to collectively track and identify the autonomic behavior of these athletes. In this context, the inverse association observed between LnRMSSD and stress perception in the present study is in line with evidence indicating that vagal suppression and the relative increase in sympathetic activity may occur in periods of greater psychophysiological load, reflecting increased allostatic cost [11,21]. Thus, the convergence between HRV, sRPE, and perceived stress reinforces the importance of a multidimensional approach to monitoring internal load, particularly in young athletes exposed to high-intensity competitive environments, in which emotional and maturational factors interact with physiological responses to training [22].

Based on current evidence, it is possible to identify a clear need for studies that adopt integrated and longitudinal approaches and that incorporate internal load metrics such as HRV, biopsychosocial context, considering the developmental phase in young athletes. In this sense, the present study aims to evaluate autonomic modulation in competitive soccer players under 17 years of age through HRV analysis with integration of subjective markers, to map the state of autonomic balance of the group, and to devise better recovery and physical readiness strategies in the context of athletic development.

2. Materials and Methods

2.1. Study Design

This study was designed as an observational longitudinal follow-up investigation, in which a cardiac variable was repeatedly assessed in the same participants over a four-week period during the competitive sports season. The objective was to assess autonomic modulation through heart rate variability (HRV) in elite youth soccer players, considering physiological and psychophysiological parameters within a real-world competitive context. All procedures were conducted in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Declaration of Helsinki and its later amendments to the World Medical Association [23]. The study was approved by the Research Ethics Committee of Santa Cecília University (CEP-Unisanta) by CAAE: 83677724.2.0000.5513, date: 8 November 2024.

2.2. Participants

The initial sample consisted of 15 male elite youth soccer players in the under-17 category of Santos Football Club (SFC). Inclusion criteria were as follows: (I) signing of the informed consent form (ICF), (II) being medically cleared to train, (III) regular participation in the main squad's training routine, and (IV) absence of a history of cardiovascular disease or risk factors for non-communicable chronic diseases (NCDs). Additionally, full attendance during the four-week data collection period was required. By the end of the study, 14 athletes met all criteria and were included in the final analysis (N = 14).

The sample was selected by convenience and represented the entire eligible under-17 squad available during the monitoring period. Similar sample sizes have been reported in previous longitudinal investigations involving elite youth athletes, reflecting the practical constraints associated with monitoring highly trained players in competitive settings.

2.3. HRV Data Collection

HRV measurements were performed once a week, setting a weekly measure for each athlete, that occurs in the morning, prior to training sessions, in a temperature-controlled environment. To minimize potential confounding influences on HRV, all assessments were performed under standardized conditions in the morning, at approximately the same time for each athlete and prior to any training activity. Participants were instructed to

avoid caffeine intake before testing, maintain their habitual hydration and dietary routines, refrain from additional strenuous physical activity, and report any factors that could affect autonomic function through a daily health and readiness questionnaire administered before each measurement [3,6].

Athletes remained seated for a one-minute stabilization period followed by a one-minute recording phase, totaling a two-minute ultra-short protocol, validated for athletic contexts [9]. The ultra-short HRV protocol was chosen because it provides a practical and time-efficient approach for monitoring autonomic modulation in applied sports settings, particularly in team-sport environments where athletes are assessed repeatedly within limited daily routines. Previous evidence has shown that ultra-short-term HRV recordings, especially RMSSD-derived indices, can provide valid information regarding parasympathetic modulation and are sensitive to training-related changes in athletes [9]. In this context, the use of a brief seated protocol allowed for standardized morning assessments with minimal disruption to the training schedule, while maintaining ecological validity and feasibility for longitudinal monitoring in elite youth soccer players. By reducing the duration of the assessment and standardizing the body position, time of day, and pre-training conditions, this protocol may also help minimize measurement variability and improve the consistency of repeated HRV recordings across the monitoring period.

Data were collected using a Polar H9 chest strap monitor (model 1 W; Polar Electro Oy, Kempele, Finland). Recordings were analyzed using Kubios HRV® Scientific Lite software (version 4.1.2.1; Kubios Oy, Kuopio, Finland), with standard automatic artifact correction filters. The primary metric extracted was the root mean square of successive differences (RMSSD), which was subsequently log-transformed (LnRMSSD) to correct data skewness and meet statistical assumptions.

2.4. Psychophysiological Questionnaires

Before each training session, the athletes completed a standardized health and readiness questionnaire to identify potential confounding factors (e.g., caffeine consumption, emotional state). The Perceived Recovery Status (PRS) scale was applied, covering four domains, (I) sleep duration (hours), (II) level of perceived stress, (III) mood state, and (IV) general perception of recovery assessed using the same questionnaire upon arrival at the training center. All items were scored using an ordinal scale from 0 to 10 [21,24]. Important to note that, perceived stress was assessed using a 0–10 ordinal scale, in which higher scores represented greater levels of perceived psychophysiological stress.

In addition, after completion of the training session, before leaving the training center, athletes reported their perceived exertion for the session, the Perceived Exertion Assessment (sRPE) scale (modified Borg scale, 0–10) was applied at the end of each training session, with 0 indicating no exertion and 10 indicating maximal exertion [25].

2.5. Physical and Anthropometric Assessments

Body mass was measured using a previously calibrated digital scale (G-Life, model CA7000, São Paulo, Brazil), with a precision of 0.1 kg, while height was obtained using a fixed stadiometer (Sanny, model ES2030; American Medical do Brasil Ltda., São Bernardo do Campo, Brazil), with a precision of 0.1 cm. During the evaluations, the athletes remained barefoot and wearing light clothing, positioned in an orthostatic posture, with the body aligned and the head oriented in the Frankfort plane.

Maximum oxygen uptake was estimated using the Yo-Yo Intermittent Recovery Test—Level 1 (Yo-Yo IR1), which is widely used to assess aerobic fitness in intermittent sports. The test consists of repeated 20 m runs back and forth, performed at a progressively increasing speed, controlled by sound signals emitted by standardized audio. Between each stage,

athletes perform a 10 s active recovery period, covering 5 m. The test was conducted on the field and ended when the athlete failed to reach the marking line for two consecutive times in the time determined by the sound signal or when voluntary exhaustion occurred. The total distance covered was recorded and used to estimate VO_2max using the equation proposed for the test [26,27].

Body composition was estimated by measuring skinfolds using a scientific skinfold compass (Sanny®, model AD1010-1; American Medical do Brasil Ltda., São Bernardo do Campo, Brazil). The measurements were always performed by the same trained evaluator, following the standardized recommendations for anthropometric evaluation. Seven skinfolds were evaluated: pectoral, middle axillary, triceps, subscapular, abdominal, suprailiac and thigh. Each measurement was performed on the right side of the body and repeated twice, considering the mean of the values obtained. Body density was estimated from the equation proposed by Jackson and Pollock [28] for seven skinfolds, and later converted into body fat percentage using the Siri equation.

2.6. Training Protocol

During the observation period, players engaged in approximately 20 field training sessions, with an average of 4 to 5 sessions per week, and 10 strength training sessions, corresponding to approximately 3 sessions per week. Strength training sessions were organized as complementary neuromuscular training and included exercises with external overload and field-transfer actions, such as sled work, resisted traction exercises, free squats, lunges, hip thrusts, plyometric exercises, changes in direction, and tasks aimed at transferring power to soccer-specific actions.

Field sessions followed the regular training routine of the club and involved technical–tactical and physical contents, including small-sided games, possession games, formal games, positional exercises, and conditioning tasks integrated into the team's playing model. The intensity and content of the sessions varied according to their proximity to the official match within the competitive microcycle. For this purpose, match day was defined as MD, and the days preceding the match were classified according to their distance from the game, such as MD-4, MD-3, MD-2, and MD-1. In general, MD-4 sessions emphasized strength, mechanical load, changes in direction, and short games. On MD-3, speed stimuli and field-based power training were prioritized. On MD-2, sessions were characterized by moderate volume and lower relative intensity, whereas on MD-1, tactical activities predominated, with lower volume, lower neuromuscular demand, and greater focus on immediate match preparation.

The athletes also participated in three official matches, with an average playing time of 60.2 $\pm$ 27.06 min per match, reflecting the competitive exposure during the study period. Data collection took place at the beginning of the competitive season, after approximately 40 days of pre-season preparation, which allowed for a prior period of physiological adaptation before the monitoring phase. Although detailed quantitative external-load variables were not included in the present analysis, the characterization of session content, microcycle organization, and competitive exposure provides important contextual information for interpreting the HRV and perceptual-marker responses observed during the study. It is important to emphasize that all athletes had been involved in competitive soccer for at least 5 years.

2.7. Statistical Analysis

Statistical analyses were performed using IBM SPSS Statistics version 25.0 (IBM Corp., Armonk, NY, USA). Data distribution was assessed using the Shapiro–Wilk test because it

is recommended for small sample sizes. Continuous variables are presented as the mean ± standard deviation.

Pearson's product-moment correlation coefficient was used to examine the associations between LnRMSSD and the perceptual variables (PRS, sRPE, and perceived stress) after verification of the assumptions of normality and linearity.

Longitudinal changes in RMSSD and LnRMSSD across the four monitoring time points were analyzed using the Friedman test, a non-parametric alternative for repeated measures, because the repeated observations did not fully satisfy the assumptions required for repeated-measures ANOVA. Statistical significance was set at $p < 0.05$.

3. Results

A total of fourteen elite youth soccer met the inclusion criteria and were included in the study sample. The athletes had a mean age of 16.2 ± 0.42 years. Anthropometric characteristics and indirectly assessed cardiorespiratory fitness are presented in Table 1.

Table 1. Player characteristics at the beginning of the season.

Variable	Mean	Standard Deviation
Body mass (kg)	71.0	±7.5
Height (m)	1.76	±0.07
VO_2max (mL·kg^{-1}·min^{-1})	51.5	±3.31
Body fat (%)	6.6	±1.78

Data are presented as mean ± standard deviation.

3.1. Longitudinal Changes in Heart Rate Variability

Longitudinal analysis was performed to examine HRV modulation across the four sequential monitoring time points, represented in Table 2.

Table 2. Longitudinal changes in heart rate variability across the four monitoring time points.

Variable	Time 1	Time 2	Time 3	Time 4	X^2	Df	p
RMSSD mean (ms)	68.78 ± 33.6	70.85 ± 46.6	67.29 ± 36.0	67.01 ± 43.8	1.71	3	0.634
LnRMSSD mean	4.09 ± 0.63	4.04 ± 0.7	4.02 ± 0.7	4.16 ± 0.7	2.74	3	0.434

Data are presented as the mean ± standard deviation. Temporal differences were examined using the Friedman test. Statistic significance was set at $p < 0.05$.

Over the four-week longitudinal monitoring period, the athletes exhibited no statistically significant temporal difference across the four monitoring time points, as evidenced by mean values of LnRMSSD = 4.06 ± 0.68 and RMSSD = 68.78 ± 33.62 ms (Figure 1A), suggesting the predominance of vagal tone and a favorable recovery status, despite a competitive calendar involving high training and match density. These values fell within the reference ranges reported in the literature for youth athletes under competitive conditions, reflecting appropriate physiological adaptation to the internal load demands.

Internal training load was further characterized using perceptual subjective measures collected daily. The group exhibited mean values of Session Rating of Perceived Exertion (sRPE) = 6.36 ± 1.41, Perceived Recovery Status (PRS) = 7.66 ± 1.60, Stress = 6.96 ± 1.96, and Mood = 7.35 ± 2.12. These results indicate a generally favorable perception of recovery and mood, accompanied by moderate levels of perceived stress during the monitoring period.

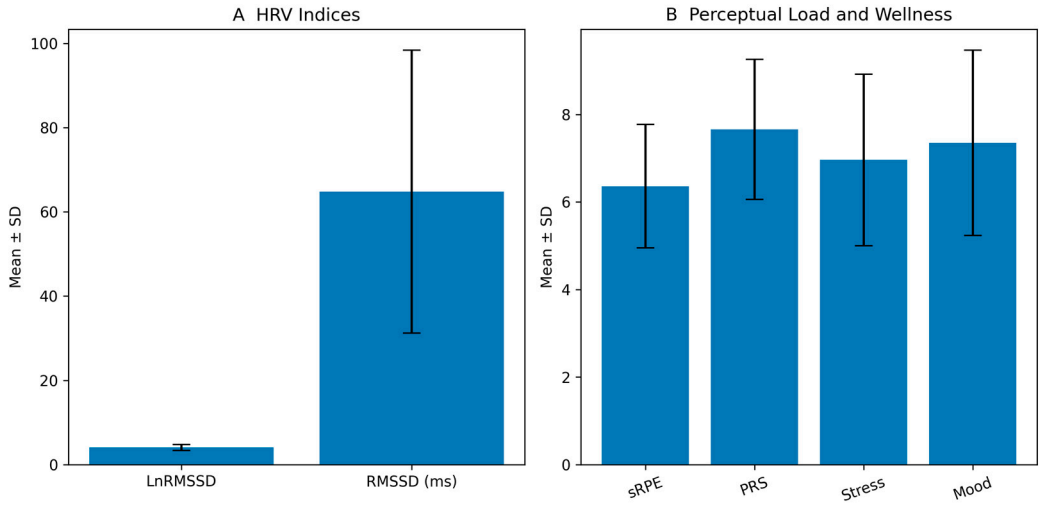


Figure 1. Physiological and perceptual monitoring indices in elite youth players. (**A**) Heart rate variability markers expressed as the natural logarithm of the root mean square of successive differences (LnRMSSD) and absolute RMSSD (ms), representing parasympathetic modulation and short-term beat-to-beat variability, respectively. (**B**) Perceptual load and wellness indicators including session rating of perceived exertion (sRPE), perceived recovery status (PRS), perceived stress, and mood state. Bars represent mean values and error bars denote standard deviation (±SD), indicating inter-individual variability.

3.2. Correlations Between LnRMSSD and Subjective Variables

In Figure 2A, we demonstrate the correlations between LnRMSSD and PRS. A moderate positive correlation was observed (r = 0.61), indicating that higher recovery perception scores were associated with greater parasympathetic activity.

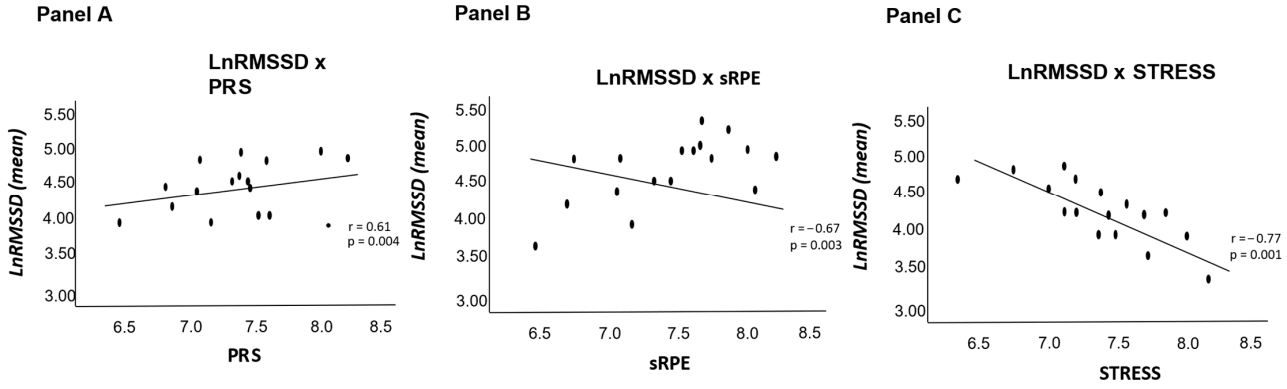


Figure 2. (**A**) shows the correlation between LnRMSSD and mean values of perceived recovery status (PRS), demonstrating a moderate positive association (r = 0.61). (**B**) illustrates the relationship between LnRMSSD and mean values of Rating of Perceived Exertion (sRPE), with a moderate-to-strong negative correlation (r = −0.67). (**C**) presents the association between LnRMSSD and mean values of perceived stress, revealing a strong correlation (r = −0.77).

As illustrated in Figure 2B, LnRMSSD showed a moderate inverse association with sRPE indicating that lower parasympathetic modulation was related to higher perceived training load (r = −0.67).

Finally, LnRMSSD was moderately and inversely correlated with perceived stress (r = −0.77), indicating that athletes presenting reduced vagal-related HRV tended to report greater psychophysiological stress (Figure 2C).

These associations underscore the coherence between physiological and subjective indicators of load, highlighting the value of integrative monitoring strategies in youth athletic development.

4. Discussion

The present study aimed to investigate the autonomic modulation of youth soccer players in a competitive context through the analysis of heart rate variability (HRV), using the time-domain parameter RMSSD and its logarithmic transformation (LnRMSSD). The method employed followed the ultra-short protocol, validated in sports settings [7,9], allowing for efficient data collection compatible with the dynamic and time-limited routines of team sports.

The findings revealed average values of LnRMSSD and RMSSD, consistent with the expected range for well-recovered young athletes in high-demand training periods. These values align with previous studies indicating that LnRMSSD values between 3.5 and 4.1 are indicative of parasympathetic predominance and functional recovery capacity in intermittent sports athletes [8,19,29].

No statistically significant temporal changes in RMSSD or LnRMSSD, even in the face of physical and psychological demands imposed by competition, suggests good physiological adaptation and efficient recovery capacity among the athletes. This reinforces the relevance of monitoring HRV as a sensitive and non-invasive biomarker capable of assisting in individual load control and in anticipating maladaptation scenarios or increased injury risk [11,21,30].

In addition, the integrated analysis between physiological indicators and subjective measures of perceived recovery (PRS) and perceived exertion (sRPE) revealed moderate and consistent correlations, indicating coherence between autonomic responses and athletes' perceptions of internal load. These findings suggest that variations in cardiac autonomic modulation were accompanied by perceptual responses related to recovery status and training load. Previous evidence indicates that subjective scales, when systematically applied, are sensitive to psychophysiological changes induced by training and competition and can reliably reflect athletes' functional status [24]. In this context, combining heart rate variability indices, particularly those related to parasympathetic modulation, with perceptual measures such as PRS and sRPE may provide a more contextualized interpretation of internal load and training readiness in team-sport athletes [21,31]. This integrated approach may be particularly relevant in youth athletes, in whom emotional, psychosocial, and maturational factors can modulate autonomic responses to training while simultaneously influencing perceived exertion and recovery [22].

The observed inverse relationship between LnRMSSD and sRPE supports the use of sRPE as a valid marker of internal training load [24]. Specifically, higher perceived training loads were associated with lower cardiac autonomic modulation, as reflected by reduced LnRMSSD values. This pattern may indicate a shift toward greater sympathetic predominance and reduced parasympathetic activity, which has previously been associated with increased physiological stress and allostatic load in athletes [11,21].

Moreover, the inverse correlation observed between LnRMSSD and perceived stress reinforces the role of psychological factors in modulating cardiac autonomic control. These results corroborate the multidimensional perspective proposed by recent research, emphasizing that HRV should be interpreted considering contextual variables, including psychological, behavioral, and training-related aspects [11,21,22]. This relationship may be explained by the fact that psychological stress directly influences central autonomic regulatory mechanisms, promoting greater sympathetic activation and reducing parasympathetic influence on the heart. As LnRMSSD is an index predominantly associated with

vagal modulation, lower values observed in athletes reporting higher levels of perceived stress may indicate a transient reduction in autonomic recovery capacity and decreased physiological flexibility in response to competitive demands.

In youth athletes, this response may be particularly relevant, as emotional factors, performance pressure, routine instability, and psychosocial demands inherent to athletic development can amplify stress responses and further disrupt sympathovagal balance. Therefore, the inverse correlation observed between LnRMSSD and perceived stress reinforces the notion that psychological state should not be interpreted as an isolated variable, but rather as a factor capable of directly modulating cardiac autonomic control and, consequently, influencing heart rate variability measures used to assess recovery status and readiness to perform.

These findings are consistent with the literature, which demonstrates that subjective scales applied in a systematic way are sensitive to detecting psychophysiological changes induced by training and competition, in addition to being associated with physiological indicators of internal load [24]. In addition, investigations that combined HRV measures with perceptual indicators demonstrate that the integration between vagal indices, such as the LnRMSSD, and subjective variables allows for a more comprehensive assessment of training readiness and functional status in team sports [21,31].

In the present study, an association was also observed between autonomic indicators and variables related to perceived stress, suggesting that periods of greater psychophysiological load may be accompanied by changes in autonomic modulation. Longitudinal studies conducted in young athletes subjected to intense competitive calendars indicate that periods characterized by greater training and competition demands tend to be associated with reduced parasympathetic modulation and a relative predominance of sympathetic activity, reflecting greater accumulated physiological stress [8,11]. These findings reinforce that autonomic balance is highly sensitive to competitive demands even in young populations, being influenced by both physical and psychosocial factors [32–34].

It is noteworthy that autonomic development during adolescence is marked by progressive changes in vagal modulation and sympathetic–parasympathetic interaction, influenced by the maturational stage and biological growth [35,36]. Investigations in base categories also indicate that the maturational process can modulate the autonomic response to training, changing the magnitude of heart rate variability in the face of high-intensity stimuli [29,36].

In this context, the inverse association observed between LnRMSSD and stress perception in the present study is in line with evidence indicating that vagal suppression and the relative increase in sympathetic activity may occur in periods of greater psychophysiological load, reflecting increased allostatic cost [11,21]. Thus, the convergence between HRV, sRPE, and perceived stress reinforces the importance of a multidimensional approach to monitoring internal load, particularly in young athletes exposed to high-intensity competitive environments, in which emotional and maturational factors interact with physiological responses to training [22].

From an applied perspective, the present findings support the inclusion of HRV as a complementary component of athlete monitoring systems, particularly in youth development environments. Given its practical and non-invasive nature, HRV monitoring may contribute to a more comprehensive understanding of athletes' physiological status and may assist practitioners in making informed decisions regarding training and recovery.

The limitations observed in the present study are characterized by the use of self-report-based instruments (PRS, sRPE), which, although widely validated in monitoring internal load, may be influenced by perceptual biases in their application, through momentary emotional variations and individual distortions in the interpretation of the scales. In

addition, the small sample size (N = 14) limits the generalization of the findings. The relatively short follow-up period (four weeks) may also have constrained the identification of more pronounced autonomic variations throughout different phases of the season. Finally, the absence of complementary objective measures, such as external load indicators or biochemical markers, limits the possibility of greater association of data and more robust interpretation of the observed responses. Another limitation is that biological maturation was not directly assessed. Although all participants belonged to the under-17 category, adolescence is characterized by substantial inter-individual variability in growth and maturational status, which may influence autonomic regulation, vagal modulation, and HRV responses. Therefore, the absence of objective maturational indicators, such as peak height velocity, Tanner staging, or skeletal age assessment, limits a more precise interpretation of the potential influence of biological maturation on the observed autonomic responses [35,36].

5. Conclusions

In conclusion, the results of this study suggest that the analyzed group of under-17 soccer players maintained stable autonomic modulation throughout a competitive period, with HRV values consistent with recovery and physiological balance. The correlations with subjective markers such as sRPE and PRS reinforce the relevance of integrating physiological and perceptual indicators in training monitoring.

These findings strengthen the scientific foundation for the implementation of HRV monitoring in youth soccer, highlighting its practicality, sensitivity, and potential to support decision-making in training planning. Furthermore, they emphasize the importance of adopting a multidimensional and individualized approach to promote the long-term development and health of young athletes.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

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